

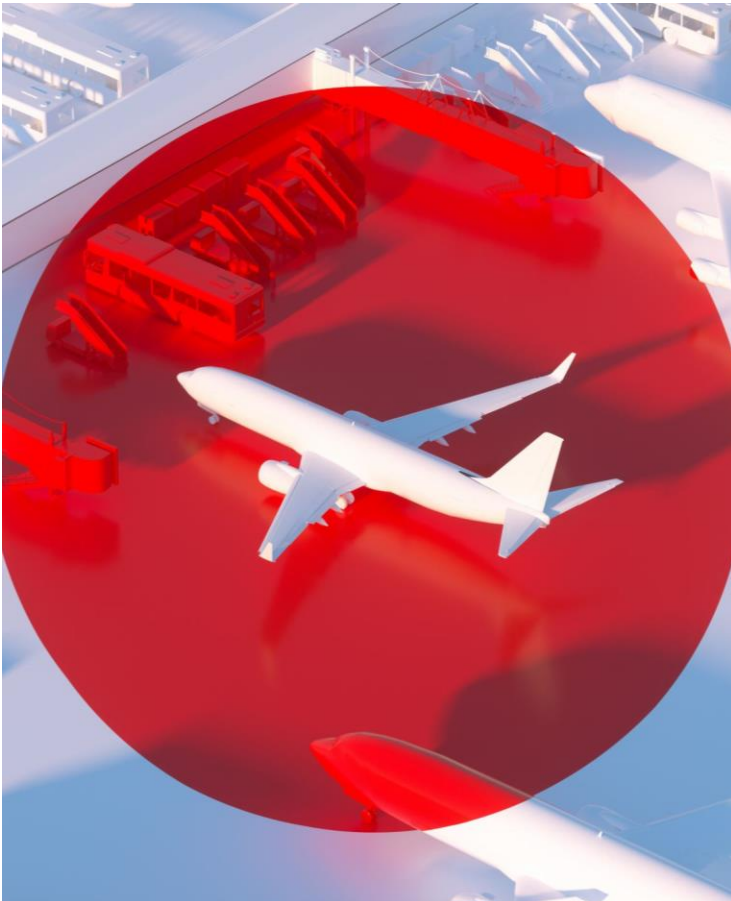
AVIATION RISK ANALYSIS

A photograph of an airplane wing in flight, viewed from the side. The wing is dark grey and extends from the left towards the right. Below the wing, a layer of white clouds is visible against a clear blue sky. The perspective is from a high altitude, looking out from the aircraft.

A comprehensive breakdown of aviation accident risks based on historical data.

By: Norman Mwapea

BUSINESS PROBLEM



High-Risk Blind Spot

A company is entering aviation without clear visibility into aircraft safety profiles- putting investment and reputation at risk.

Stakes:

- **Financial Losses:** Costly accidents or grounding of poor-performing aircraft
- **Reputational Damage:** Safety concerns could erode trust
- **Regulatory Issues:** Increased scrutiny or compliance failures

What Is Need:

- **Evidence-based model risk rankings**
- **Insights into accident trends and contributors**
- **Strategic recommendations** to guide safe aircraft procurement

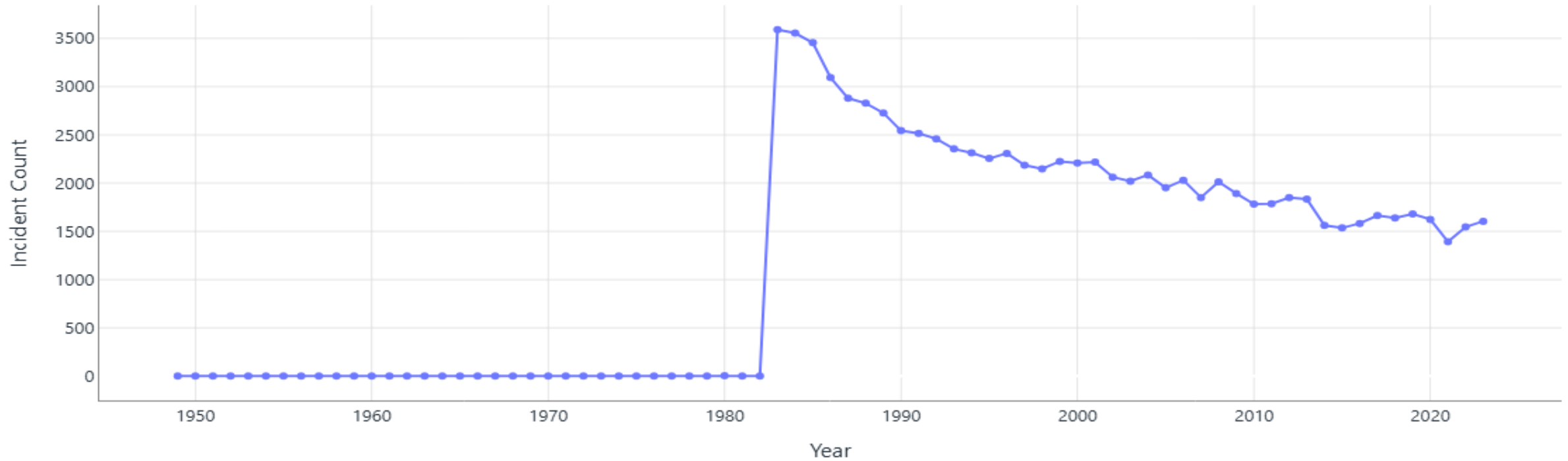
Overview of Aviation Incidents

Summarizing aviation incidents by:

1. Which aircraft manufacturers are more prone to accidents- make and model involved in accidents and their fatalities (Total Fatally injured and Total Uninjured Passengers).
2. What engine configurations of the aforementioned makes and models are involved in the highest number of accidents.
3. Which airplanes are safer- homebuilt or factory made aircrafts?



AVIATION INCIDENTS OVER TIME

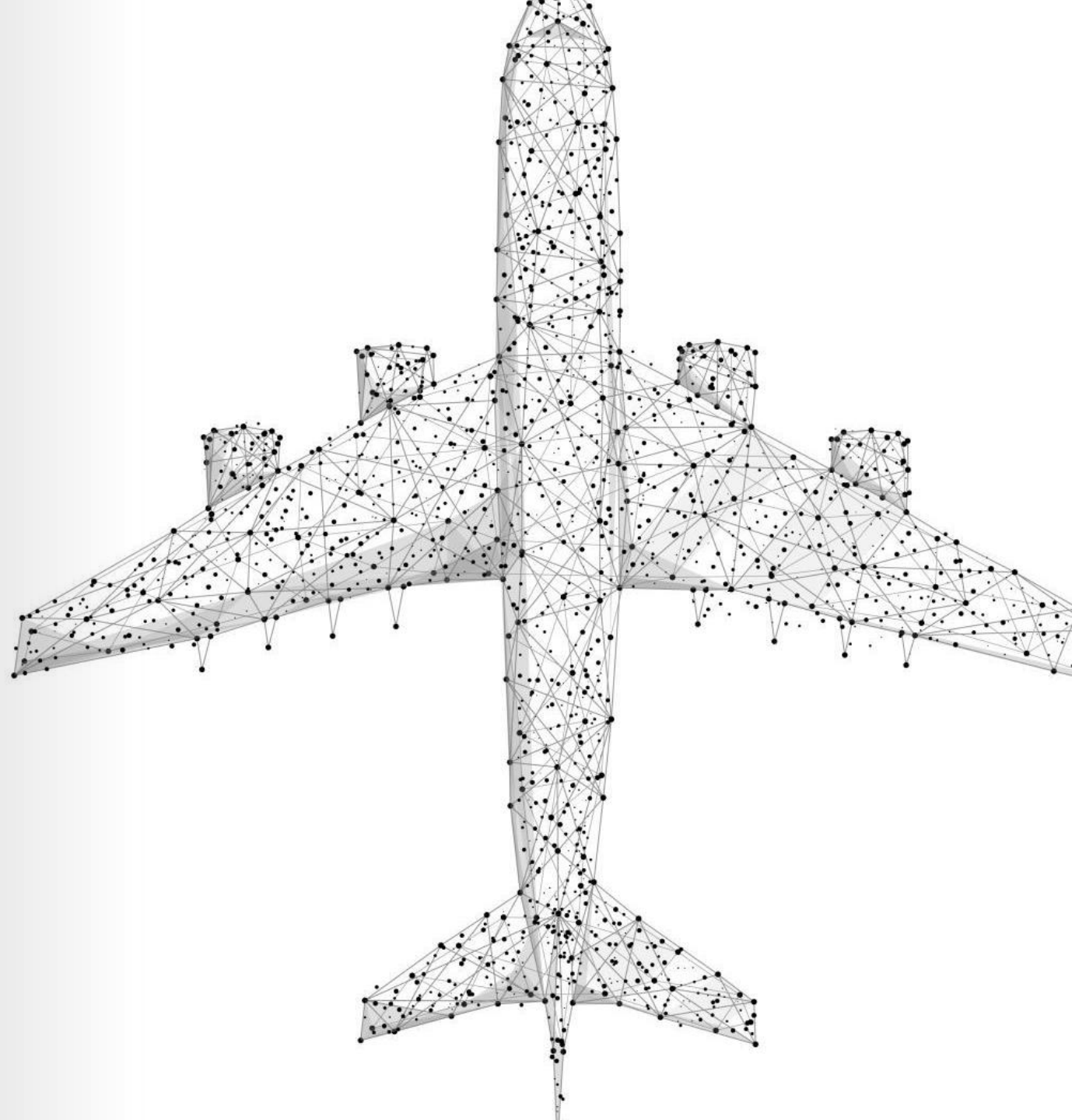


- **Peak in Early 1980s:** Aviation incidents surged sharply in 1982, possibly due to improved reporting standards or data availability.
- **Consistent Decline Since:** Steady reduction in incident counts over the past four decades, highlighting improvements in **aviation safety, technology, and regulatory enforcement**.
- **Stabilization Post-2010:** Incident rates have plateaued, indicating a matured industry- but risks still persist, especially with modern complexities.

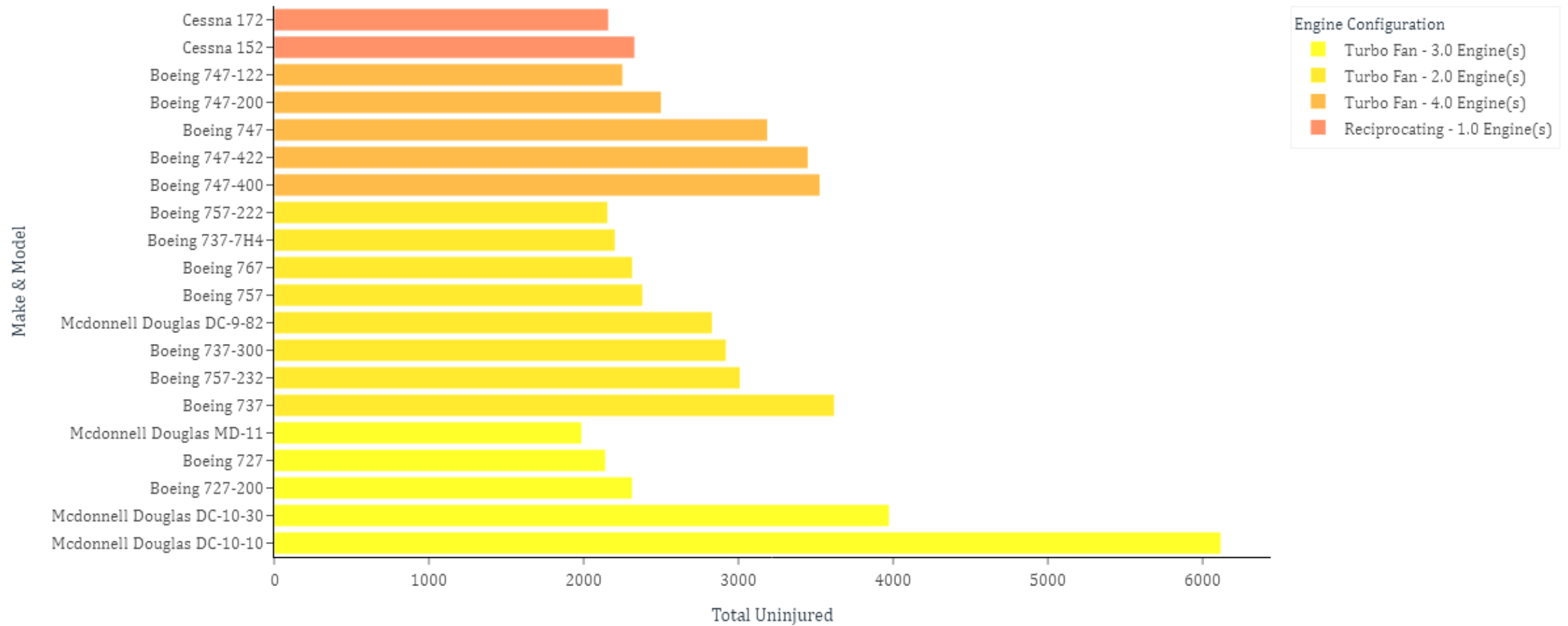
Insight: While aviation is safer than ever, selecting historically low-risk aircraft remains essential for risk-averse investment.

Types of Aircraft Involved

Analysis by aircraft make, model, and engine configuration.



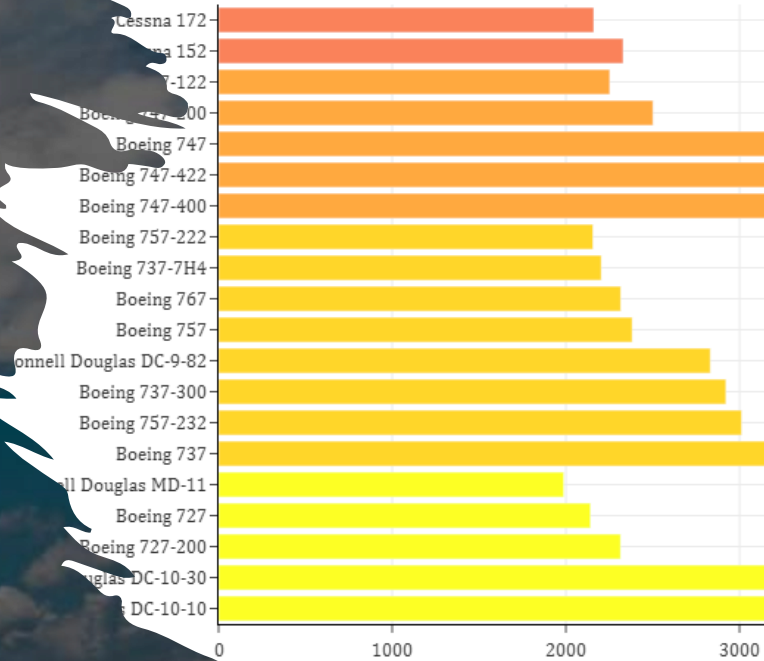
Survivability by Make & Model and Engine Configuration



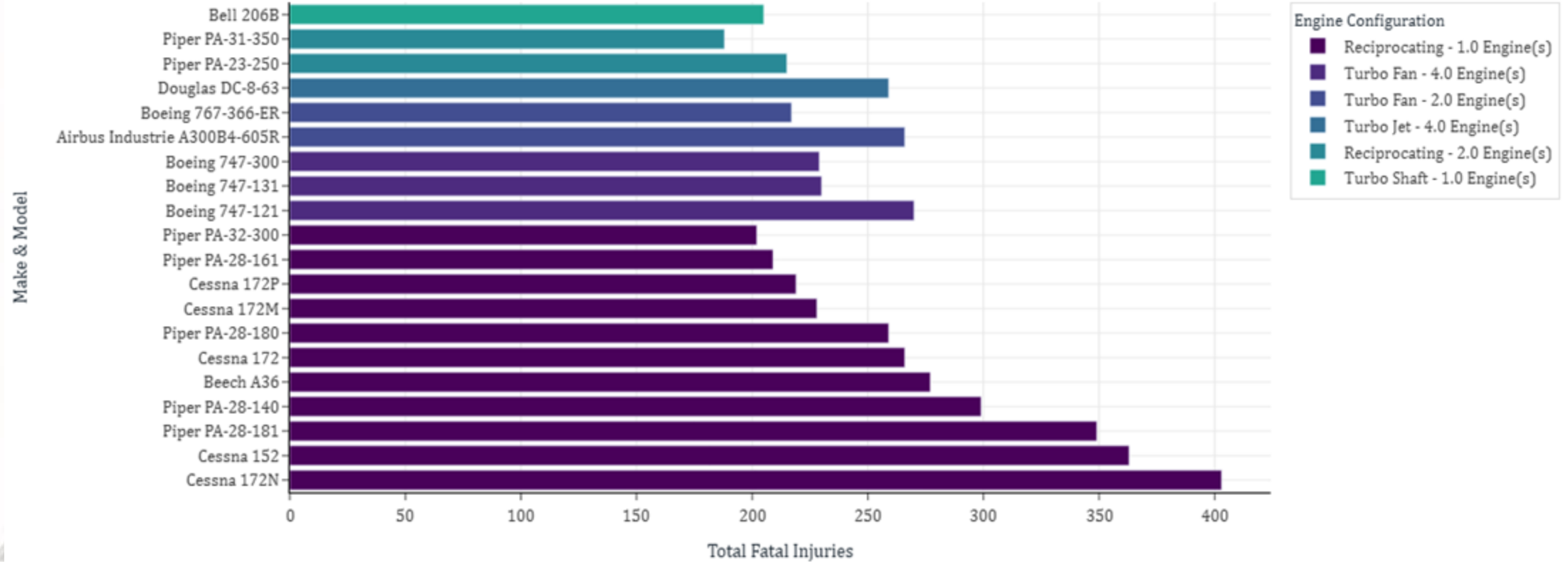
Aircraft Survivability by Make, Model, and Engine Configuration

- Aircrafts with 3 or 4 turbo fan engines consistently show better survivability.
- Planes like the **McDonnell Douglas DC-10-10** lead in total uninjured passengers, suggesting a strong link between **engine configuration** and **passenger safety**.

Survivability by Make & Model and Engine Configuration



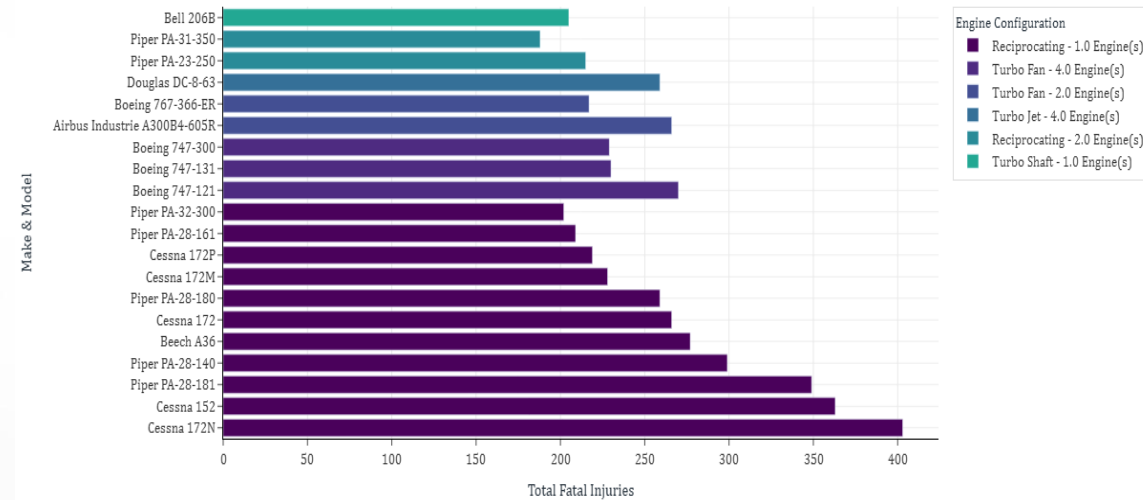
Survivability by Make & Model and Engine Configuration

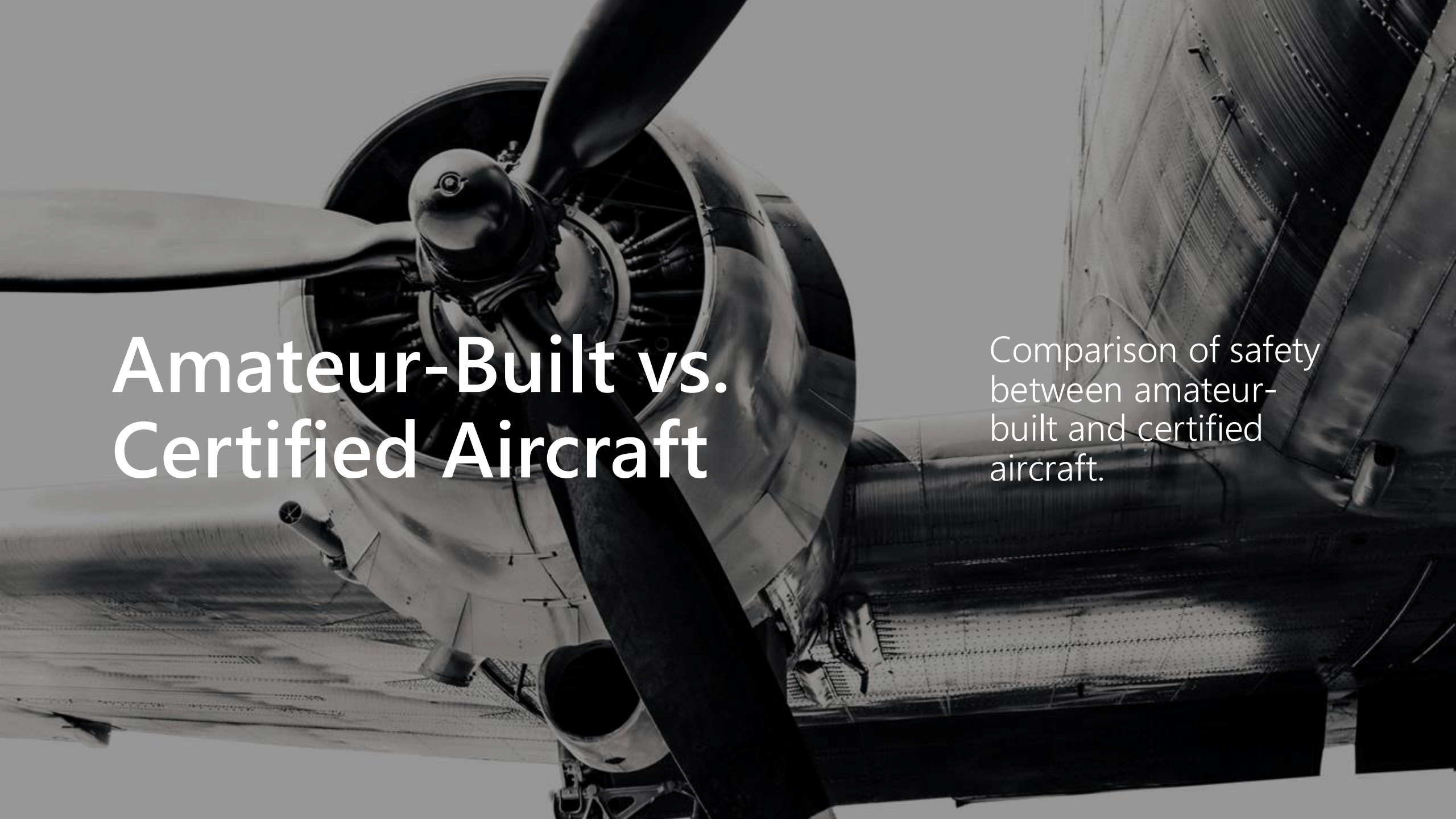


Fatalities by Aircraft Type

- The dark purple bars dominate the lower half of the chart- indicating **higher fatality counts**, especially among small aircraft like the **Cessna 172N** and **Cessna 152**.
- Meanwhile, aircraft with **multi-engine turbo configurations** consistently appear at the top with **lower fatality counts**.

Survivability by Make & Model and Engine Configuration

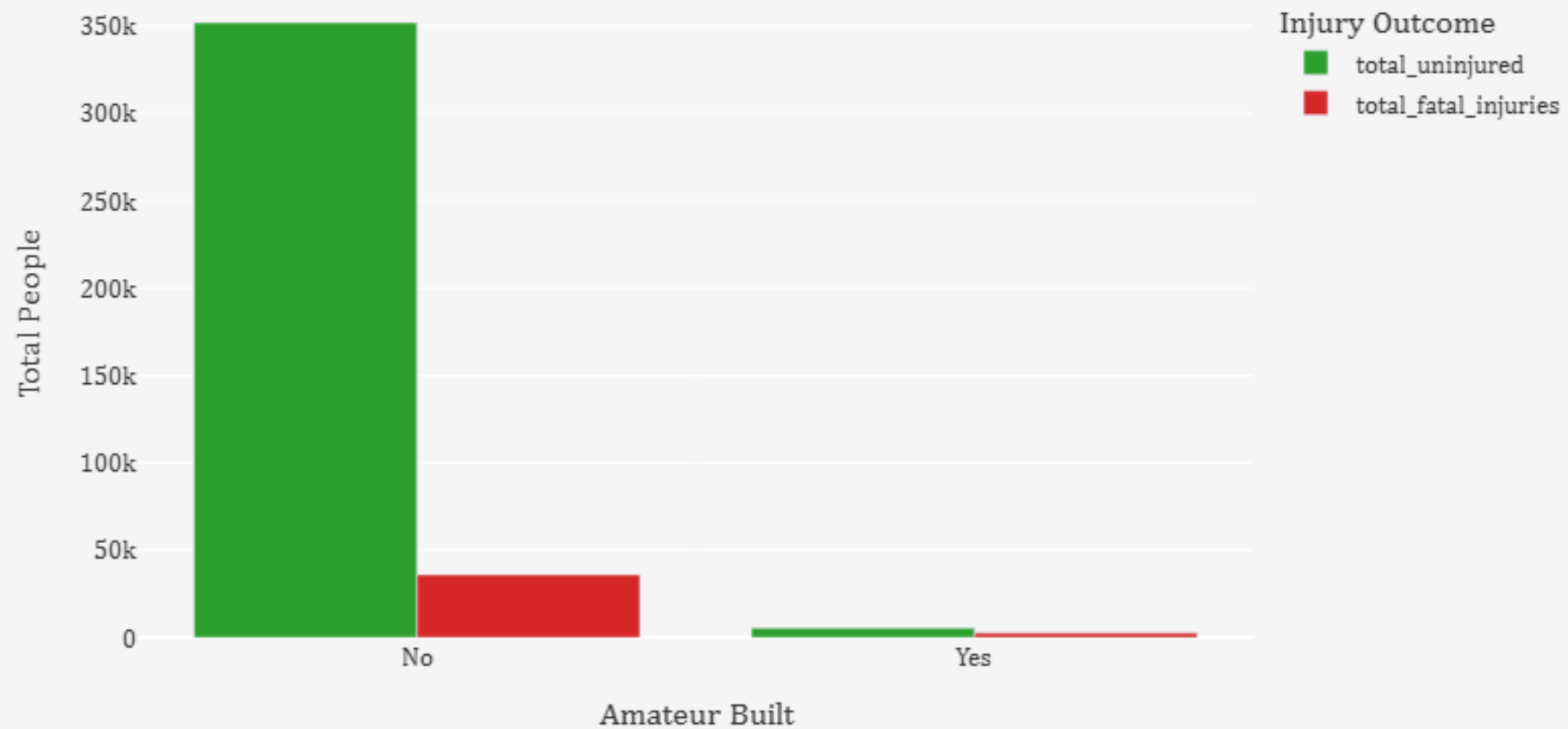




Amateur-Built vs. Certified Aircraft

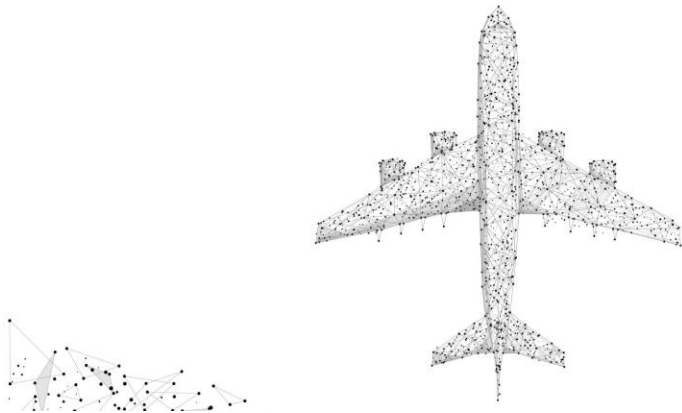
Comparison of safety
between amateur-
built and certified
aircraft.

Injuries vs Fatalities by Amateur-Built Aircraft



Is DIY Worth the Risk? Fatalities in Amateur-Built Aircraft

- While professionally built planes have far more recorded incidents they also show a **much higher rate of survivability**.
- In contrast, amateur-built aircraft have a **significantly higher fatality rate relative to their usage**—a red flag when it comes to safety.



Recommendations & Conclusion

Name _____

Signature _____

Date _____

Safest Aircraft to Add to Your Fleet

What Makes These Aircraft Safer?

- Turbo fan engines (2–3) — consistently linked to better outcomes
- Strong uninjured passenger counts
- Low relative fatality rates
- Reliable commercial performance history



Aircraft

Why It's Safe

McDonnell Douglas DC-10-10

Highest survivability; 3 turbo fan engines

McDonnell Douglas DC-10-30

Strong safety pattern; same engine config

Boeing 727

Rugged tri-engine jet with proven safety

Boeing 737-700

Lower fatalities, globally trusted short-haul jet

Boeing 757-232

Balanced design, high survivability rate

McDonnell Douglas MD-11

Advanced safety features + 3-engine design

Airbus A300B4-605R

Solid build, wide-body with lower fatality counts